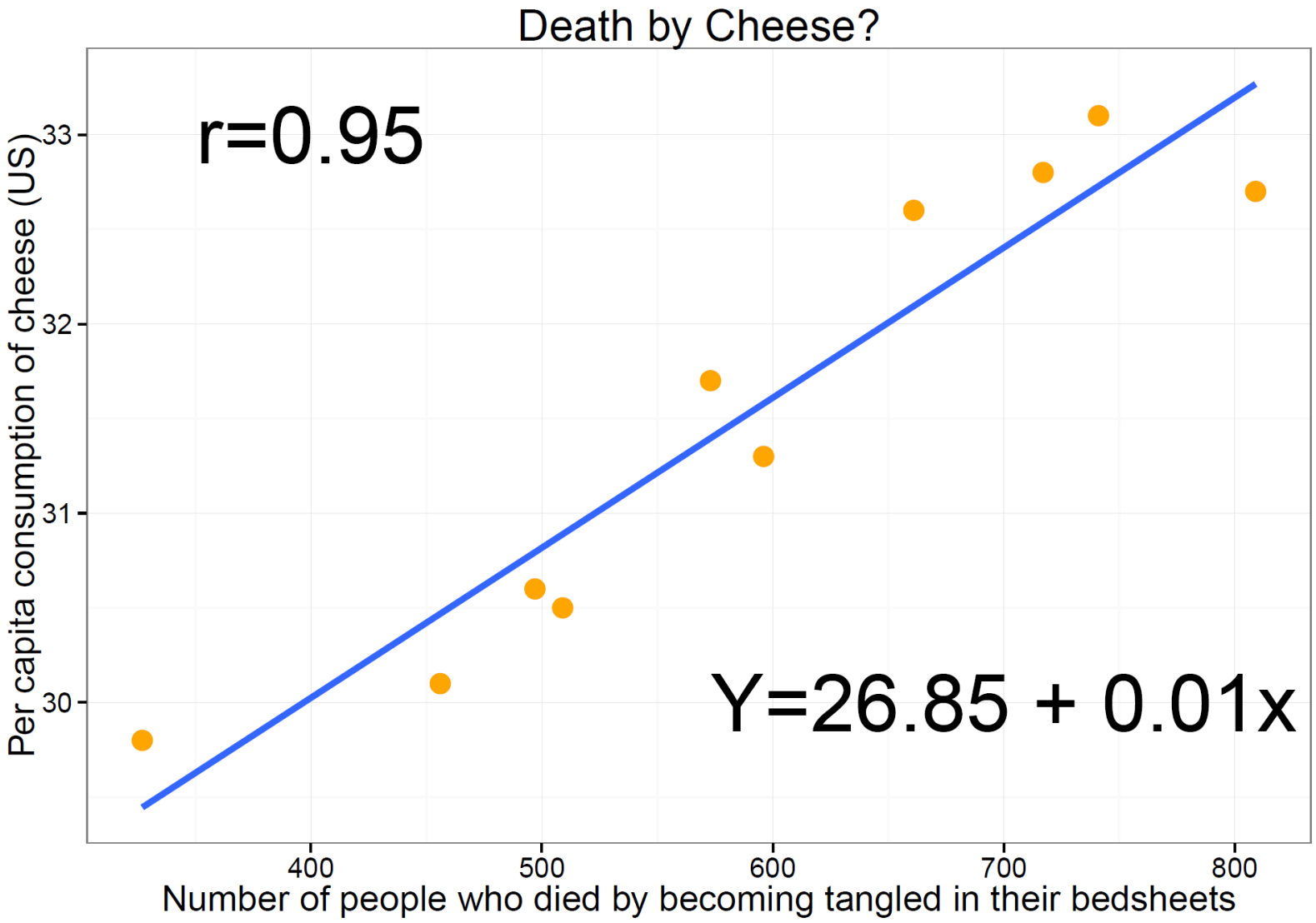
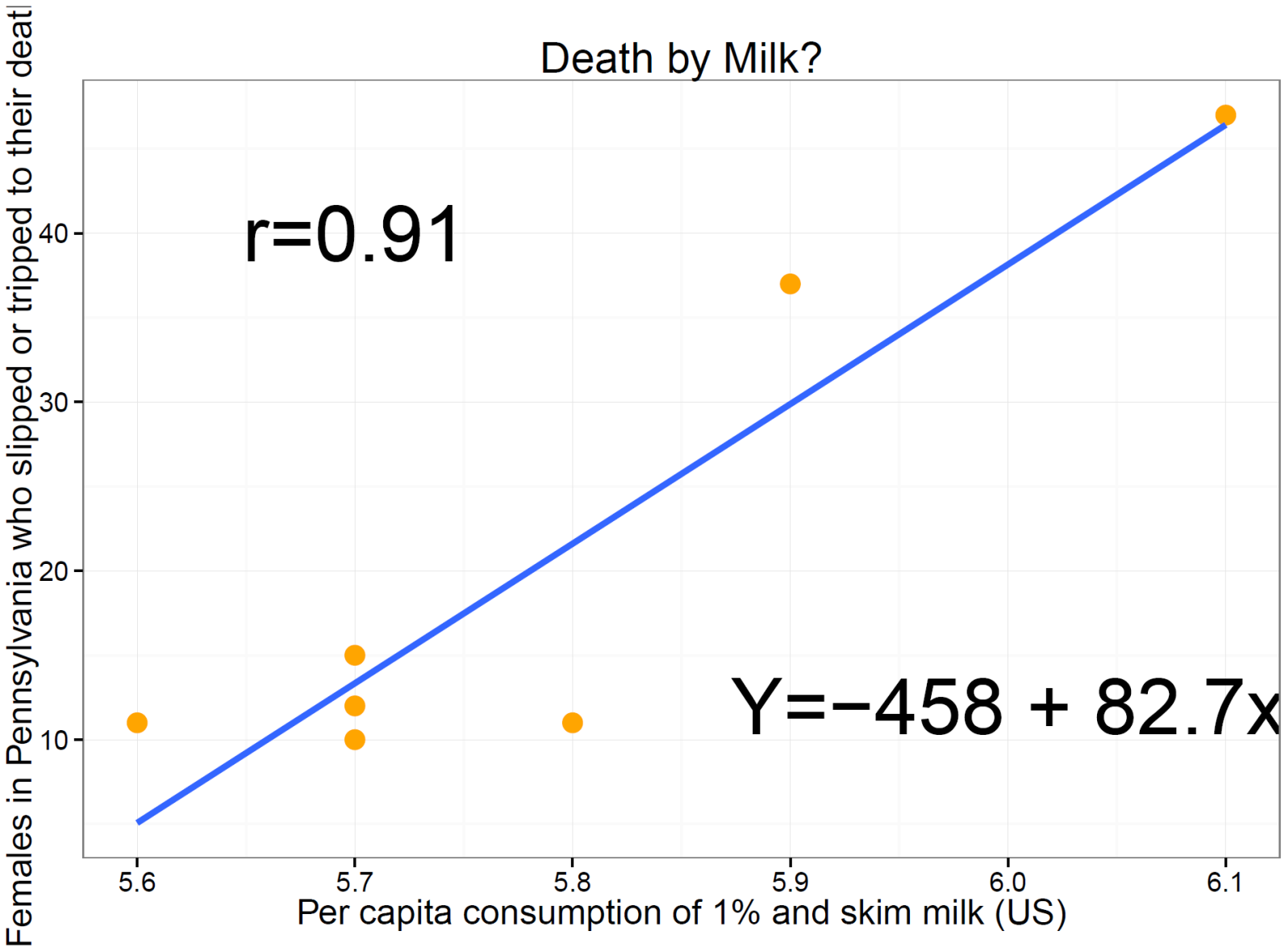
# Multivariate Regression

November 2, 2022

# Correlation Ain't Causation



# Correlation Ain't Causation



# Establishing Causation (Reminder)

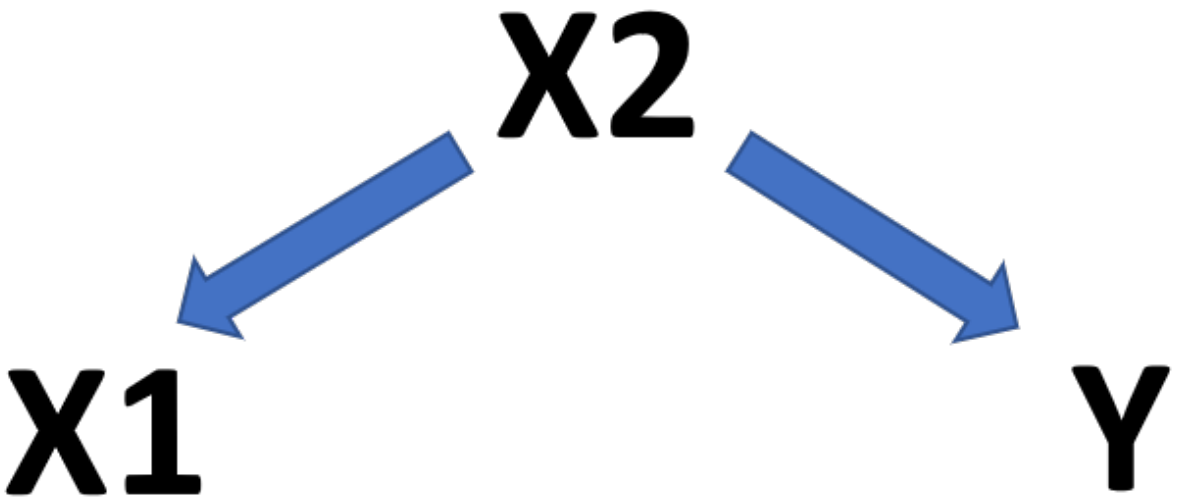
# Establishing Causation (Reminder)

1. Covariation between **X** and **Y**
2. Theory/Causal Mechanism
3. Temporal Order (**X** leads to **Y** and not the other way around)
4. All confounders ruled out

# Establishing Causation (Reminder)

1. Covariation between **X** and **Y**
2. Theory/Causal Mechanism
3. Temporal Order (**X** leads to **Y** and not the other way around)
4. All confounders ruled out
   - Random assignment to treatment or control groups
   - "Control" for other important factors

# Spurious Relationships



# Simpson's Paradox

Suppose we are interested in how legal counsel affects outcomes in criminal drunk driving cases:

| | Public Def. | Private |
|---|---|---|
| Win | 20 | 24 |
| Lose | 20 | 16 |
| Percent Win | 50% | 60% |

# Simpson's Paradox

Suppose we are interested in how legal counsel affects outcomes in criminal drunk driving cases:

| | Public Def. | Private |
|---|---|---|
| Win | 20 | 24 |
| Lose | 20 | 16 |
| Percent Win | 50% | 60% |

But what if sex is also relevant?

# Simpson's Paradox

We can control for (account for) <u>both</u> independent variables at the same time.

Men:

| | Public Def. | Private |
|---|---|---|
| Win | 12 | 3 |
| Lose | 18 | 7 |
| Percent Win | 40% | 30% |

Women:

| | Public Def. | Private |
|---|---|---|
| Win | 8 | 21 |
| Lose | 2 | 9 |
| Percent Win | 80% | 70% |

# The Multivariate World

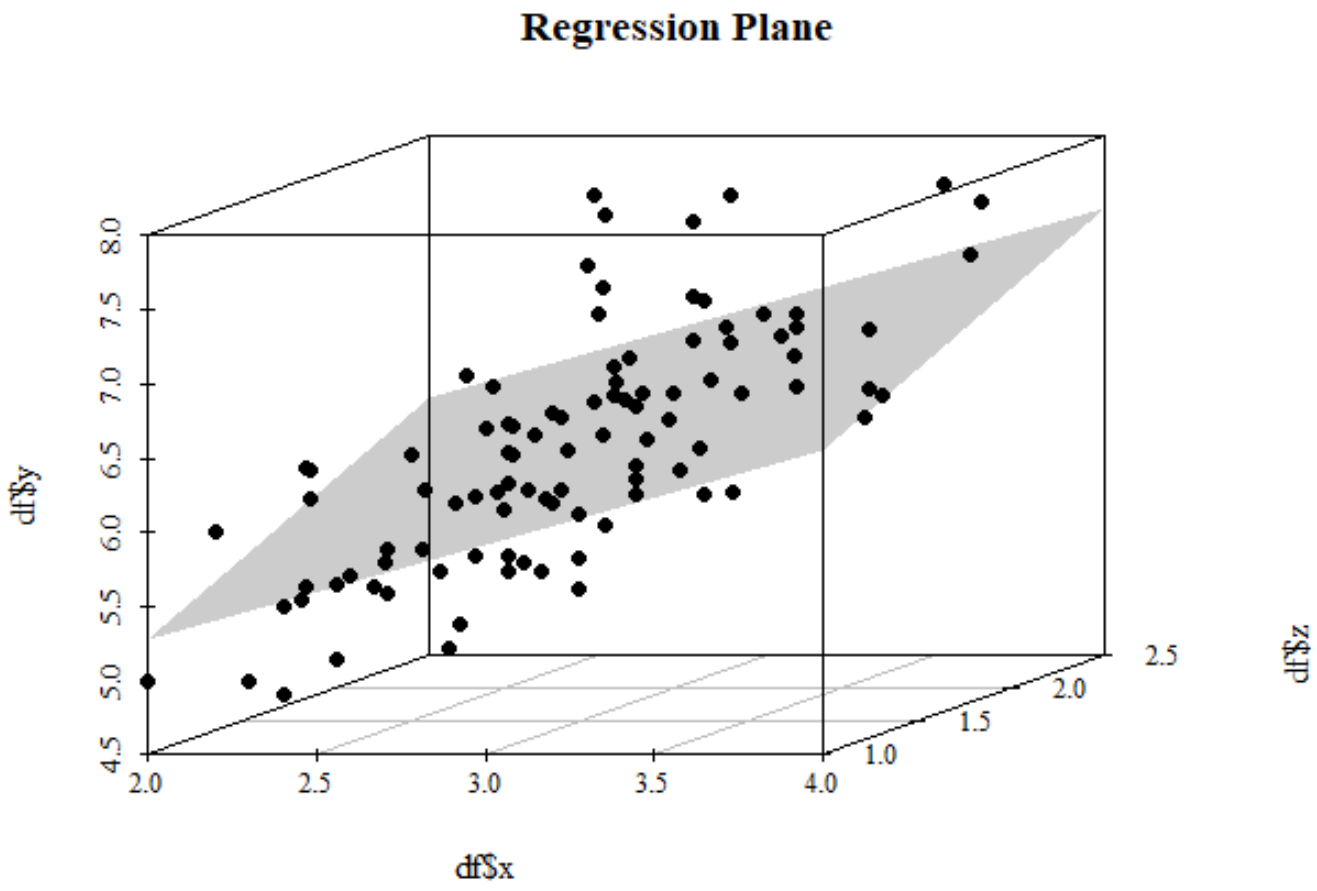
- If we do not control for the right variables, we might get the wrong results
- In a multivariate regression, we can include two or more independent variables at the same time: $X_1$, $X_2$, $X_3$, ...

# The Multivariate World

- If we do not control for the right variables, we might get the wrong results
- In a multivariate regression, we can include two or more independent variables at the same time: $X_1$, $X_2$, $X_3$, ...
- To find the effect of $X_1$ while controlling for $X_2$ (or vice versa), we estimate:

$$Y_i = \alpha + \beta_1 X_1 + \beta_2 X_2 + \epsilon_i \tag{1}$$

# Multiple Regression



| | *Dependent variable:* | | |
|---|---|---|---|
| | % Dems in Leg. | | |
| | (1) | (2) | (3) |
| % Democrat | 1.032* | | |
| | (0.252) | | |
| % Republican | | −1.518* | |
| | | (0.237) | |
| % Nonwhite | | | 0.307* |
| | | | (0.131) |
| Constant | 22.684* | 103.167* | 48.381* |
| | (8.215) | (7.600) | (3.652) |
| Observations | 49 | 49 | 49 |
| $R^2$ | 0.263 | 0.467 | 0.105 |
| Residual Std. Error | 12.944 | 11.015 | 14.269 |

| | *Dependent variable:* |
|---|---|
| | % Leg. Dems |
| % Democrat | 0.954* |
| | (0.299) |
| % Nonwhite | 0.069 |
| | (0.141) |
| Constant | 23.547* |
| | (8.468) |
| Observations | 49 |
| $R^2$ | 0.267 |
| Adjusted $R^2$ | 0.235 |
| Residual Std. Error | 13.051 (df = 46) |
| *Note:* | *p<0.05 |

# Calculating Multiple Regression

In a bivariate regression, the slope $\beta$ is estimated as:

$$\hat{\beta} = \frac{\sum[(X_i - \bar{X})(Y_i - \bar{Y})]}{\sum(X_i - \bar{X})^2} \tag{2}$$

In multiple regression, the estimate for $\beta_1$, for instance, is:

$$\hat{\beta}_1 = \frac{\sum(X_i - \hat{X}_i)(Y_i - \hat{Y}_i^*)}{\sum(X_i - \hat{X}_i)^2} \tag{3}$$

- $\hat{X}_i = X_1$ as predicted by $X_2$
- $\hat{Y}_i^* = Y$ as predicted by $X_2$

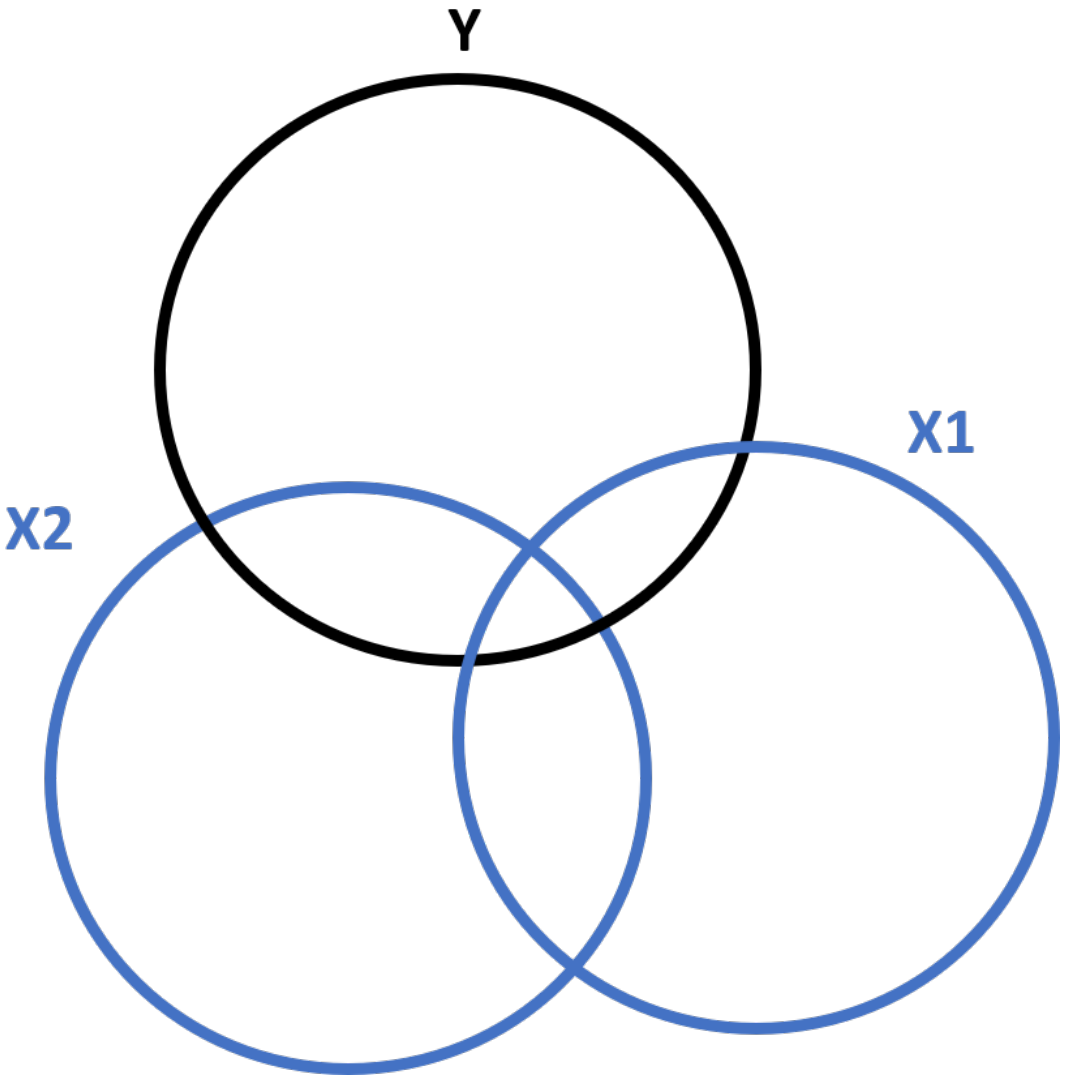
# How Does $\beta_1$ Control for Z?

$$\hat{\beta}_1 = \frac{\sum(X_i - \hat{X}_i)(Y_i - \hat{Y}_i^*)}{\sum(X_i - \hat{X}_i)^2} \tag{4}$$

- $\hat{X}_i = X_1$ as predicted by $X_2$
- $\hat{Y}_i^* = Y$ as predicted by $X_2$

*$\beta_1$ will almost always be somewhat different than in a bivariate regression!*

# Control Variables: Correlate with Y and X1



# Interstate War

**Table 10.2 Excerpts from Morrow, Siverson, and Tabares's table on the political causes of international trade**

| | A | B | C | D |
|---|---|---|---|---|
| Peaceful relations | 1.12* | — | — | 1.45* |
| | (0.22) | — | — | (0.37) |
| Democratic partners | — | 1.18* | — | 1.22* |
| | — | (0.12) | — | (0.13) |
| Alliance partners | — | — | 0.29* | −0.50* |
| | — | — | (0.03) | (0.16) |
| GNP of exporter | 0.67* | 0.57* | 0.68* | 0.56* |
| | (0.07) | (0.07) | (0.07) | (0.08) |
| $R^2$ | 0.77 | 0.78 | 0.77 | 0.78 |
| N | 2631 | 2631 | 2631 | 2631 |

Note: Other variables were estimated as a part of the regression model but were excluded from this table for ease of presentation.

Standard errors are in parentheses.

*$p < 0.05$.

# Interpreting Multiple Regression

- $\hat{\alpha}$: Expected value of Y when <u>all</u> X $= 0$
- $\hat{\beta}_k$: Expected change in Y as $X_k$ increases by one unit, **<u>all else held constant</u>**

The model can be extended to include as many explanatory variables as you want (technically), as long as $k > n$... but...

# Which Effect is Biggest?

Coefficients ($\beta$'s) are *unstandardized*, so cannot compare magnitude directly.

- Slopes are based on measurement of $X_k$
- If $X_1$ is measured in dollars (e.g., income per capita) and $X_2$ is measured in number of people (e.g., population), then coefficients not directly comparable

# Which Effect is Biggest?

Coefficients ($\beta$'s) are *unstandardized*, so cannot compare magnitude directly.

- Slopes are based on measurement of $X_k$
- If $X_1$ is measured in dollars (e.g., income per capita) and $X_2$ is measured in number of people (e.g., population), then coefficients not directly comparable

Standardized Coefficients:

$$\hat{\beta_{std.}} = \hat{\beta} \times \frac{\text{SD of X}}{\text{SD of Y}} \tag{5}$$

# Model Fit with Multiple Regression

$R^2$ and Residual SD mean the same thing! but $R^2$ isn't ideal for multiple regression. Why?

$$R^2 = \frac{b}{Y} = \frac{\text{Model Sum of Squares}}{\text{Total Sum of Squares (TSS)}} \tag{6}$$

# Adjusted $R^2$

$$R^2_{adj.} = 1 - [\frac{(1 - R^2)(n - 1)}{n - k - 1}] \tag{7}$$

Accounts for number of variables in the model

# Dummy Variables

Your Independent Variables can be categorical! To do this, include one fewer variable than you have categories.

- If you want to include a variable with values of 0 or 1 (or no and yes, etc.)
- If you have more than 2 categories, leave out one, and the omitted category is the <u>baseline</u>

# Interpreting Dummy Variables: Bivariate Regression

- $\alpha =$ average value of Y when X $= 0$
- $\alpha + \beta =$ average value of Y when X $= 1$
- $\beta =$ difference in average value of Y when X $= 0$ and X $= 1$

| | *Dependent variable:* |
|---|---|
| | % Dems in Leg. |
| Southern State | −2.627 |
| | (4.578) |
| Constant | 56.350* |
| | (2.616) |
| Observations | 49 |
| $R^2$ | 0.007 |
| Adjusted $R^2$ | −0.014 |
| Residual Std. Error | 15.029 |
| *Note:* | *p<0.05 |

# Interpreting Dummy Variables: Multiple Categories

- Coefficient is the difference between the presence of each category and the baseline

Table: Linear regression with categorical independent variables.

| | *Dependent variable:* | |
|---|---|---|
| | % Leg. Dems. | |
| Western State | −1.067 | −16.180* |
| | (5.180) | (6.015) |
| Midwestern State | −3.223 | −18.336* |
| | (5.433) | (6.235) |
| Northeastern State | 15.113* | |
| | (5.780) | |
| Southern State | | −15.113* |
| | | (5.780) |
| Constant | 53.722* | 68.836* |
| | (3.468) | (4.624) |
| Observations | 49 | 49 |
| $R^2$ | 0.190 | 0.190 |
| Adjusted $R^2$ | 0.136 | 0.136 |
| *Note:* | *p<0.05 | |

# Fitting a Multivariate Regression in R (Generally)

regression$<-$lm(formula = **y-variable** $\sim$ **x1** + **x2** + **x3**, data = **datasetname**)

summary(regression)